In [4]:
```
!pip install nlpaug==1.1.11
```

```
Collecting nlpaug==1.1.11
  Downloading nlpaug-1.1.11-py3-none-any.whl.metadata (14 kB)
Requirement already satisfied: numpy>=1.16.2 in /usr/local/lib/python3.11/dist-packages (from nlpaug==1.1.11) (1
.26.4)
Requirement already satisfied: pandas>=1.2.0 in /usr/local/lib/python3.11/dist-packages (from nlpaug==1.1.11) (2
.2.2)
Requirement already satisfied: requests>=2.22.0 in /usr/local/lib/python3.11/dist-packages (from nlpaug==1.1.11)
(2.32.3)
Requirement already satisfied: gdown>=4.0.0 in /usr/local/lib/python3.11/dist-packages (from nlpaug==1.1.11) (5.
2.0)
Requirement already satisfied: beautifulsoup4 in /usr/local/lib/python3.11/dist-packages (from gdown>=4.0.0->nlp
aug==1.1.11) (4.13.3)
Requirement already satisfied: filelock in /usr/local/lib/python3.11/dist-packages (from gdown>=4.0.0->nlpaug==1
.1.11) (3.17.0)
Requirement already satisfied: tqdm in /usr/local/lib/python3.11/dist-packages (from gdown>=4.0.0->nlpaug==1.1.1
1) (4.67.1)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.11/dist-packages (from pandas>=1
.2.0->nlpaug==1.1.11) (2.8.2)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.11/dist-packages (from pandas>=1.2.0->nlpa
ug==1.1.11) (2025.1)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.11/dist-packages (from pandas>=1.2.0->nl
paug==1.1.11) (2025.1)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.11/dist-packages (from request
s>=2.22.0->nlpaug==1.1.11) (3.4.1)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.11/dist-packages (from requests>=2.22.0->n
lpaug==1.1.11) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.11/dist-packages (from requests>=2.2
2.0->nlpaug==1.1.11) (2.3.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.11/dist-packages (from requests>=2.2
2.0->nlpaug==1.1.11) (2025.1.31)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.11/dist-packages (from python-dateutil>=2.8.2-
>pandas>=1.2.0->nlpaug==1.1.11) (1.17.0)
Requirement already satisfied: soupsieve>1.2 in /usr/local/lib/python3.11/dist-packages (from beautifulsoup4->gd
own>=4.0.0->nlpaug==1.1.11) (2.6)
Requirement already satisfied: typing-extensions>=4.0.0 in /usr/local/lib/python3.11/dist-packages (from beautif
ulsoup4->gdown>=4.0.0->nlpaug==1.1.11) (4.12.2)
Requirement already satisfied: PySocks!=1.5.7,>=1.5.6 in /usr/local/lib/python3.11/dist-packages (from requests[
socks]->gdown>=4.0.0->nlpaug==1.1.11) (1.7.1)
Downloading nlpaug-1.1.11-py3-none-any.whl (410 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 410.5/410.5 kB 9.7 MB/s eta 0:00:00
Installing collected packages: nlpaug
Successfully installed nlpaug-1.1.11
```

In [5]:
```
# Import libraries
import pandas as pd
import numpy as np
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from nltk.stem import WordNetLemmatizer
import re
from sklearn.model_selection import train_test_split, KFold
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, confusion_matrix, roc_curve
from imblearn.over_sampling import SMOTE
import matplotlib.pyplot as plt
import seaborn as sns
import tensorflow as tf
from tensorflow.keras.preprocessing.text import Tokenizer
from tensorflow.keras.preprocessing.sequence import pad_sequences
from tensorflow.keras.models import Model
from tensorflow.keras.layers import Input, Embedding, LSTM, Bidirectional, Dense, Dropout, Layer
from tensorflow.keras.callbacks import EarlyStopping
import nlpaug.augmenter.word as naw  # For text augmentation
import os
import requests
import zipfile
from google.colab import drive
```

dataset --> https://cseweb.ucsd.edu/~jmcauley/datasets/amazon_v2/

In [6]:
```
nltk.download('averaged_perceptron_tagger_eng')
```

```
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data]     /root/nltk_data...
[nltk_data]   Unzipping taggers/averaged_perceptron_tagger_eng.zip.
```

Out[6]:
```
True
```

In [7]:
```
# Download NLTK resources
nltk.download('punkt_tab')
```

```
nltk.download('stopwords')
nltk.download('wordnet')
```

```
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
[nltk_data] Downloading package wordnet to /root/nltk_data...
```

Out[7]: True

In [8]:
```
from google.colab import drive
import pandas as pd

# Mount Google Drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

In [ ]:
```
# Define the path to your CSV file
file_path = '/content/drive/My Drive/Digital_Music_5.json'
```

In [ ]:
```
# Read the JSON file with lines=True
try:
    df = pd.read_json(file_path, lines=True)  # Added lines=True for JSON Lines format
    print("JSON file loaded successfully!")
    print(df.head())  # Display first few rows to verify

    # Convert to CSV
    df.to_csv('/content/drive/My Drive/Digital_Music_5.csv', index=False)
    print("Converted to CSV and saved!")
except FileNotFoundError:
    print(f"Error: File not found at {file_path}")
except ValueError as e:
    print(f"Error: Invalid JSON format - {str(e)}")
except Exception as e:
    print(f"An unexpected error occurred: {str(e)}")
```

```
JSON file loaded successfully!
   overall  vote  verified   reviewTime      reviewerID        asin  \
0        5   3.0      True   06 3, 2013  A2TYZ821XXK2YZ  3426958910
1        5   NaN      True  10 11, 2014  A3OFSREZADFUDY  3426958910
2        5   NaN      True  02 11, 2014  A2VAMODP8M77NG  3426958910
3        4   3.0     False   12 7, 2013   AAKSLZ9IDTEH0  3426958910
4        5   NaN      True  06 12, 2016  A3OH43OZJLKI09  5557706259

                      style       reviewerName  \
0  {'Format:': ' Audio CD'}            Garrett
1  {'Format:': ' Audio CD'}                 Ad
2  {'Format:': ' Audio CD'}             JTGabq
3  {'Format:': ' Audio CD'}  john F&#039;n doe
4  {'Format:': ' Audio CD'}  melinda a goodman

                                         reviewText  \
0  This is awesome to listen to, A must-have for ...
1                                               bien
2  It was great to hear the old stuff again and I...
3  well best of's are a bit poison normally but t...
4  What can I say? This is Casting Crowns!!!This ...

                                           summary  unixReviewTime image
0                                     Slayer Rules!      1370217600   NaN
1                                        Five Stars      1412985600   NaN
2                        SLAYER!!!!!!!!!!!!!!!!!!!!!      1392076800   NaN
3  slayer greatest hits! you mean everything righ...      1386374400   NaN
4                     This is a good, blessing filled      1465689600   NaN
Converted to CSV and saved!
```

In [9]:
```
df = pd.read_csv("/content/drive/My Drive/Digital_Music_5.csv")
```

In [10]:
```
print("Dataset Info:")
print(df.info())
print("\nFirst 5 rows:")
print(df.head())
```

```
Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 169781 entries, 0 to 169780
Data columns (total 12 columns):
 #   Column          Non-Null Count   Dtype
---  ------          --------------   -----
 0   overall         169781 non-null  int64
 1   vote            7611 non-null    float64
 2   verified        169781 non-null  bool
 3   reviewTime      169781 non-null  object
 4   reviewerID      169781 non-null  object
 5   asin            169781 non-null  object
 6   style           157989 non-null  object
 7   reviewerName    169763 non-null  object
 8   reviewText      169613 non-null  object
 9   summary         169720 non-null  object
 10  unixReviewTime  169781 non-null  int64
 11  image           182 non-null     object
dtypes: bool(1), float64(1), int64(2), object(8)
memory usage: 14.4+ MB
None

First 5 rows:
   overall  vote  verified   reviewTime      reviewerID        asin  \
0        5   3.0      True   06 3, 2013  A2TYZ821XXK2YZ  3426958910
1        5   NaN      True  10 11, 2014  A3OFSREZADFUDY  3426958910
2        5   NaN      True  02 11, 2014  A2VAMODP8M77NG  3426958910
3        4   3.0     False   12 7, 2013   AAKSLZ9IDTEH0  3426958910
4        5   NaN      True  06 12, 2016  A3OH43OZJLKI09  5557706259

                     style       reviewerName  \
0  {'Format:': ' Audio CD'}            Garrett
1  {'Format:': ' Audio CD'}                 Ad
2  {'Format:': ' Audio CD'}             JTGabq
3  {'Format:': ' Audio CD'}  john F&#039;n doe
4  {'Format:': ' Audio CD'}  melinda a goodman

                                        reviewText  \
0  This is awesome to listen to, A must-have for ...
1                                               bien
2  It was great to hear the old stuff again and I...
3  well best of's are a bit poison normally but t...
4  What can I say? This is Casting Crowns!!!This ...

                                        summary  unixReviewTime image
0                                  Slayer Rules!      1370217600   NaN
1                                     Five Stars      1412985600   NaN
2                       SLAYER!!!!!!!!!!!!!!!!!!!!      1392076800   NaN
3  slayer greatest hits! you mean everything righ...      1386374400   NaN
4                    This is a good, blessing filled      1465689600   NaN
```

In [11]:
```
df.drop(columns=['image'], inplace=True)
```

In [12]:
```
df.head()
```

Out[12]:

	overall	vote	verified	reviewTime	reviewerID	asin	style	reviewerName	reviewText	summary
0	5	3.0	True	06 3, 2013	A2TYZ821XXK2YZ	3426958910	{'Format:': ' Audio CD'}	Garrett	This is awesome to listen to, A must-have for ...	Slayer Rules!
1	5	NaN	True	10 11, 2014	A3OFSREZADFUDY	3426958910	{'Format:': ' Audio CD'}	Ad	bien	Five Stars
2	5	NaN	True	02 11, 2014	A2VAMODP8M77NG	3426958910	{'Format:': ' Audio CD'}	JTGabq	It was great to hear the old stuff again and I...	SLAYER!!!!!!!!!!!!!!!!!!!!
3	4	3.0	False	12 7, 2013	AAKSLZ9IDTEH0	3426958910	{'Format:': ' Audio CD'}	john F'n doe	well best of's are a bit poison normally but t...	slayer greatest hits! you mean everything righ...
4	5	NaN	True	06 12, 2016	A3OH43OZJLKI09	5557706259	{'Format:': ' Audio CD'}	melinda a goodman	What can I say? This is Casting Crowns!!!This ...	This is a good, blessing filled

In [13]:
```
df.shape
```

Out[13]: (169781, 11)

In [14]:
```
# Clean dataset
df = df[['overall', 'reviewText']].dropna()
print(f"\nDataset size after cleaning: {df.shape}")
```

```
Dataset size after cleaning: (169613, 2)
```

In [15]:
```
# Binary sentiment: 1-3 = Negative (0), 4-5 = Positive (1)
df['sentiment'] = df['overall'].apply(lambda x: 1 if x >= 4 else 0)
print("\nSentiment distribution (before balancing):")
print(df['sentiment'].value_counts())
```

```
Sentiment distribution (before balancing):
sentiment
1    158825
0     10788
Name: count, dtype: int64
```

In [16]:
```
# Advanced preprocessing
lemmatizer = WordNetLemmatizer()
stop_words = set(stopwords.words('english'))
```

In [17]:
```
def preprocess_text(text):
    """Advanced preprocessing: lowercase, clean, tokenize, lemmatize, remove stopwords."""
    text = str(text).lower()
    text = re.sub(r'[^a-zA-Z\s]', '', text)
    text = re.sub(r'\s+', ' ', text).strip()
    tokens = word_tokenize(text)
    tokens = [lemmatizer.lemmatize(word) for word in tokens if word not in stop_words and len(word) > 2]
    return ' '.join(tokens)
```

In [18]:
```
df['cleaned_text'] = df['reviewText'].apply(preprocess_text)
print("\nSample of cleaned text:")
print(df[['reviewText', 'cleaned_text']].head())
```

```
Sample of cleaned text:
                                          reviewText  \
0  This is awesome to listen to, A must-have for ...
1                                               bien
2  It was great to hear the old stuff again and I...
3  well best of's are a bit poison normally but t...
4  What can I say? This is Casting Crowns!!!This ...

                                        cleaned_text
0  awesome listen musthave slayer fanssadly neede...
1                                               bien
2  great hear old stuff like new stuff recommend ...
3  well best ofs bit poison normally bad pretty g...
4        say casting crownsthis good blessing filled
```

In [19]:
```
df[['reviewText', 'cleaned_text']].head()
```

Out[19]:

	reviewText	cleaned_text
0	This is awesome to listen to, A must-have for ...	awesome listen musthave slayer fanssadly neede...
1	bien	bien
2	It was great to hear the old stuff again and I...	great hear old stuff like new stuff recommend ...
3	well best of's are a bit poison normally but t...	well best ofs bit poison normally bad pretty g...
4	What can I say? This is Casting Crowns!!!This ...	say casting crownsthis good blessing filled

In [20]:
```
# Data augmentation with nlpaug (synonym replacement)
aug = naw.SynonymAug(aug_p=0.3)  # 30% chance to replace words with synonyms
df_aug = df.copy()
```

In [21]:
```
def safe_augment(text):
    """Safely augment text, handling empty or invalid outputs."""
    if not text or len(text.strip()) == 0:  # Check for empty input
        return text
    augmented = aug.augment(text)
    # Handle both list and string outputs from augment()
    if isinstance(augmented, list):
        return augmented[0] if augmented else text  # Return first item or original if empty
    return augmented  # Return s
```

In [22]:
```
df_aug['cleaned_text'] = df_aug['cleaned_text'].apply(safe_augment)
df = pd.concat([df, df_aug], ignore_index=True)  # Double dataset size with augmented data
print(f"\nDataset size after augmentation: {df.shape}")
```

```
Dataset size after augmentation: (339226, 4)
```

In [23]:
```
# Tokenization and padding
max_words = 20000  # Larger vocab for richer representation
max_len = 200      # Longer sequences for more context
tokenizer = Tokenizer(num_words=max_words)
tokenizer.fit_on_texts(df['cleaned_text'])
sequences = tokenizer.texts_to_sequences(df['cleaned_text'])
X = pad_sequences(sequences, maxlen=max_len)
y = df['sentiment'].values
```

In [24]:
```
# Balance classes with SMOTE
smote = SMOTE(random_state=42)
X_balanced, y_balanced = smote.fit_resample(X, y)
print(f"\nBalanced dataset size: {X_balanced.shape}, Sentiment distribution: {np.bincount(y_balanced)}")
```

```
Balanced dataset size: (635300, 200), Sentiment distribution: [317650 317650]
```

In [25]:
```
# Split into train, val, test
X_temp, X_test, y_temp, y_test = train_test_split(X_balanced, y_balanced, test_size=0.2, random_state=42)
X_train, X_val, y_train, y_val = train_test_split(X_temp, y_temp, test_size=0.25, random_state=42)
print(f"Train: {X_train.shape}, Validation: {X_val.shape}, Test: {X_test.shape}")
```

```
Train: (381180, 200), Validation: (127060, 200), Test: (127060, 200)
```

In [26]:
```
# Load GloVe 100d embeddings (download if not in Drive)
glove_path = '/content/drive/My Drive/glove.6B.100d.txt'
embedding_dim = 100  # Changed to 100d
embedding_index = {}
```

In [27]:
```
if not os.path.exists(glove_path):
    print(f"GloVe file not found at {glove_path}. Downloading...")
    url = 'http://nlp.stanford.edu/data/glove.6B.zip'
    glove_zip_path = '/content/glove.6B.zip'
    response = requests.get(url)
    with open(glove_zip_path, 'wb') as f:
        f.write(response.content)
    with zipfile.ZipFile(glove_zip_path, 'r') as zip_ref:
        zip_ref.extract('glove.6B.100d.txt', '/content')
    glove_path = '/content/glove.6B.100d.txt'
    print("GloVe file downloaded and extracted.")
```

```
GloVe file not found at /content/drive/My Drive/glove.6B.100d.txt. Downloading...
GloVe file downloaded and extracted.
```

In [28]:
```
print("Loading GloVe 100d embeddings...")
with open(glove_path, 'r', encoding='utf-8') as f:
    for line in f:
        values = line.split()
        word = values[0]
        coefs = np.asarray(values[1:], dtype='float32')
        embedding_index[word] = coefs
```

```
Loading GloVe 100d embeddings...
```

In [29]:
```
# Create embedding matrix
word_index = tokenizer.word_index
embedding_matrix = np.zeros((max_words, embedding_dim))
for word, i in word_index.items():
    if i < max_words:
        embedding_vector = embedding_index.get(word)
        if embedding_vector is not None:
            embedding_matrix[i] = embedding_vector
print(f"Embedding matrix shape: {embedding_matrix.shape}")
```

```
Embedding matrix shape: (20000, 100)
```

DEEP LEARNING MODEL IMPLEMENTATION

Goal: Advanced Bidirectional LSTM with custom Attention, manual hyperparameter tuning, gradient clipping

In [31]:
```
# Define custom Attention layer
class CustomAttention(Layer):
    def __init__(self, **kwargs):
        super(CustomAttention, self).__init__(**kwargs)

    def build(self, input_shape):
        self.W = self.add_weight(name='attention_weight', shape=(input_shape[-1], 1),
                                 initializer='random_normal', trainable=True)
        self.b = self.add_weight(name='attention_bias', shape=(input_shape[1], 1),
                                 initializer='zeros', trainable=True)
        super(CustomAttention, self).build(input_shape)
```

```
    def call(self, inputs):
        # Attention mechanism: e = tanh(W * h + b), alpha = softmax(e)
        e = tf.tanh(tf.tensordot(inputs, self.W, axes=1) + self.b)
        alpha = tf.nn.softmax(e, axis=1)
        context = inputs * alpha
        return tf.reduce_sum(context, axis=1)
```

In [32]:
```
# Define model function
def create_model(lstm_units=128, dropout_rate=0.5):
    """Create Bidirectional LSTM with custom Attention."""
    inputs = Input(shape=(max_len,))
    x = Embedding(max_words, embedding_dim, weights=[embedding_matrix], trainable=False)(inputs)
    x = Bidirectional(LSTM(lstm_units, return_sequences=True))(x)
    x = CustomAttention()(x)
    x = Dropout(dropout_rate)(x)
    x = Dense(64, activation='relu')(x)
    x = Dropout(dropout_rate)(x)
    outputs = Dense(1, activation='sigmoid')(x)
    model = Model(inputs, outputs)
    optimizer = tf.keras.optimizers.Adam(clipnorm=1.0)  # Gradient clipping
    model.compile(optimizer=optimizer, loss='binary_crossentropy', metrics=['accuracy'])
    return model
```

In [33]:
```
# Manual hyperparameter tuning with k-fold CV
param_candidates = [
    {'lstm_units': 64, 'dropout_rate': 0.3},
    {'lstm_units': 128, 'dropout_rate': 0.3},
    {'lstm_units': 64, 'dropout_rate': 0.5},
    {'lstm_units': 128, 'dropout_rate': 0.5}
]
kfold = KFold(n_splits=3, shuffle=True, random_state=42)
best_score = 0
best_params = None
```

In [34]:
```
print("\nPerforming 3-fold CV for hyperparameter tuning (optimized for speed)...")
for param_idx, params in enumerate(param_candidates, 1):
    print(f"\nEvaluating Parameter Set {param_idx}/{len(param_candidates)}: {params}")
    fold_scores = []
    for fold_idx, (train_idx, val_idx) in enumerate(kfold.split(X_train), 1):
        print(f"  Fold {fold_idx}/3 starting...")
        X_fold_train, X_fold_val = X_train[train_idx], X_train[val_idx]
        y_fold_train, y_fold_val = y_train[train_idx], y_train[val_idx]
        model = create_model(**params)
        print(f"    Training model with {len(X_fold_train)} samples...")
        model.fit(X_fold_train, y_fold_train, epochs=2, batch_size=64, verbose=1,  # Reduced epochs, increased b
                  validation_data=(X_fold_val, y_fold_val),
                  callbacks=[EarlyStopping(monitor='val_loss', patience=2, restore_best_weights=True)])
        score = model.evaluate(X_fold_val, y_fold_val, verbose=0)[1]  # Accuracy
        fold_scores.append(score)
        print(f"  Fold {fold_idx}/3 completed. Validation Accuracy: {score:.4f}")
    avg_score = np.mean(fold_scores)
    print(f"Parameter Set {param_idx} completed. Average CV Accuracy: {avg_score:.4f}")
    if avg_score > best_score:
        best_score = avg_score
        best_params = params

print(f"\nBest Params: {best_params}, Best CV Accuracy: {best_score:.4f}")
```

```
Performing 3-fold CV for hyperparameter tuning (optimized for speed)...

Evaluating Parameter Set 1/4: {'lstm_units': 64, 'dropout_rate': 0.3}
  Fold 1/3 starting...
    Training model with 254120 samples...
Epoch 1/2
3971/3971 ━━━━━━━━━━━━━━━━━━━━ 108s 25ms/step - accuracy: 0.7351 - loss: 0.5132 - val_accuracy: 0.8219 - val_los
s: 0.3886
Epoch 2/2
3971/3971 ━━━━━━━━━━━━━━━━━━━━ 136s 25ms/step - accuracy: 0.8459 - loss: 0.3453 - val_accuracy: 0.8574 - val_los
s: 0.3228
  Fold 1/3 completed. Validation Accuracy: 0.8574
  Fold 2/3 starting...
    Training model with 254120 samples...
Epoch 1/2
3971/3971 ━━━━━━━━━━━━━━━━━━━━ 103s 25ms/step - accuracy: 0.7343 - loss: 0.5202 - val_accuracy: 0.8400 - val_los
s: 0.3536
Epoch 2/2
3971/3971 ━━━━━━━━━━━━━━━━━━━━ 142s 25ms/step - accuracy: 0.8459 - loss: 0.3484 - val_accuracy: 0.8563 - val_los
s: 0.3226
  Fold 2/3 completed. Validation Accuracy: 0.8563
  Fold 3/3 starting...
    Training model with 254120 samples...
Epoch 1/2
3971/3971 ━━━━━━━━━━━━━━━━━━━━ 104s 25ms/step - accuracy: 0.7164 - loss: 0.5482 - val_accuracy: 0.8182 - val_los
```

```
s: 0.3964
Epoch 2/2
3971/3971 ━━━━━━━━━━━━━━━━━━━━ 142s 25ms/step - accuracy: 0.8242 - loss: 0.3830 - val_accuracy: 0.8530 - val_los
s: 0.3356
  Fold 3/3 completed. Validation Accuracy: 0.8530
Parameter Set 1 completed. Average CV Accuracy: 0.8556

Evaluating Parameter Set 2/4: {'lstm_units': 128, 'dropout_rate': 0.3}
  Fold 1/3 starting...
    Training model with 254120 samples...
Epoch 1/2
3971/3971 ━━━━━━━━━━━━━━━━━━━━ 121s 30ms/step - accuracy: 0.7429 - loss: 0.5077 - val_accuracy: 0.8487 - val_los
s: 0.3413
Epoch 2/2
3971/3971 ━━━━━━━━━━━━━━━━━━━━ 142s 30ms/step - accuracy: 0.8551 - loss: 0.3291 - val_accuracy: 0.8679 - val_los
s: 0.3028
  Fold 1/3 completed. Validation Accuracy: 0.8679
  Fold 2/3 starting...
    Training model with 254120 samples...
Epoch 1/2
3971/3971 ━━━━━━━━━━━━━━━━━━━━ 123s 30ms/step - accuracy: 0.7047 - loss: 0.5551 - val_accuracy: 0.8338 - val_los
s: 0.3657
Epoch 2/2
3971/3971 ━━━━━━━━━━━━━━━━━━━━ 142s 30ms/step - accuracy: 0.8443 - loss: 0.3493 - val_accuracy: 0.8605 - val_los
s: 0.3124
  Fold 2/3 completed. Validation Accuracy: 0.8605
  Fold 3/3 starting...
    Training model with 254120 samples...
Epoch 1/2
3971/3971 ━━━━━━━━━━━━━━━━━━━━ 123s 30ms/step - accuracy: 0.7251 - loss: 0.5276 - val_accuracy: 0.8454 - val_los
s: 0.3500
Epoch 2/2
3971/3971 ━━━━━━━━━━━━━━━━━━━━ 140s 30ms/step - accuracy: 0.8511 - loss: 0.3394 - val_accuracy: 0.8671 - val_los
s: 0.3024
  Fold 3/3 completed. Validation Accuracy: 0.8671
Parameter Set 2 completed. Average CV Accuracy: 0.8652

Evaluating Parameter Set 3/4: {'lstm_units': 64, 'dropout_rate': 0.5}
  Fold 1/3 starting...
    Training model with 254120 samples...
Epoch 1/2
3971/3971 ━━━━━━━━━━━━━━━━━━━━ 98s 24ms/step - accuracy: 0.6975 - loss: 0.5650 - val_accuracy: 0.8136 - val_loss
: 0.4013
Epoch 2/2
3971/3971 ━━━━━━━━━━━━━━━━━━━━ 98s 25ms/step - accuracy: 0.8298 - loss: 0.3815 - val_accuracy: 0.8527 - val_loss
: 0.3325
  Fold 1/3 completed. Validation Accuracy: 0.8527
  Fold 2/3 starting...
    Training model with 254120 samples...
Epoch 1/2
3971/3971 ━━━━━━━━━━━━━━━━━━━━ 101s 25ms/step - accuracy: 0.7018 - loss: 0.5639 - val_accuracy: 0.8215 - val_los
s: 0.3915
Epoch 2/2
3971/3971 ━━━━━━━━━━━━━━━━━━━━ 142s 25ms/step - accuracy: 0.8259 - loss: 0.3869 - val_accuracy: 0.8526 - val_los
s: 0.3316
  Fold 2/3 completed. Validation Accuracy: 0.8526
  Fold 3/3 starting...
    Training model with 254120 samples...
Epoch 1/2
3971/3971 ━━━━━━━━━━━━━━━━━━━━ 102s 25ms/step - accuracy: 0.7105 - loss: 0.5598 - val_accuracy: 0.8254 - val_los
s: 0.3907
Epoch 2/2
3971/3971 ━━━━━━━━━━━━━━━━━━━━ 95s 24ms/step - accuracy: 0.8296 - loss: 0.3769 - val_accuracy: 0.8501 - val_loss
: 0.3404
  Fold 3/3 completed. Validation Accuracy: 0.8501
Parameter Set 3 completed. Average CV Accuracy: 0.8518

Evaluating Parameter Set 4/4: {'lstm_units': 128, 'dropout_rate': 0.5}
  Fold 1/3 starting...
    Training model with 254120 samples...
Epoch 1/2
3971/3971 ━━━━━━━━━━━━━━━━━━━━ 121s 30ms/step - accuracy: 0.7068 - loss: 0.5537 - val_accuracy: 0.8359 - val_los
s: 0.3661
Epoch 2/2
3971/3971 ━━━━━━━━━━━━━━━━━━━━ 118s 30ms/step - accuracy: 0.8405 - loss: 0.3660 - val_accuracy: 0.8556 - val_los
s: 0.3242
  Fold 1/3 completed. Validation Accuracy: 0.8556
  Fold 2/3 starting...
    Training model with 254120 samples...
Epoch 1/2
3971/3971 ━━━━━━━━━━━━━━━━━━━━ 122s 30ms/step - accuracy: 0.7326 - loss: 0.5245 - val_accuracy: 0.8442 - val_los
s: 0.3497
Epoch 2/2
```

```
3971/3971 ━━━━━━━━━━━━━━━━━━━━ 141s 30ms/step - accuracy: 0.8490 - loss: 0.3451 - val_accuracy: 0.8639 - val_los
s: 0.3121
  Fold 2/3 completed. Validation Accuracy: 0.8639
  Fold 3/3 starting...
    Training model with 254120 samples...
Epoch 1/2
3971/3971 ━━━━━━━━━━━━━━━━━━━━ 121s 30ms/step - accuracy: 0.7153 - loss: 0.5470 - val_accuracy: 0.8320 - val_los
s: 0.3706
Epoch 2/2
3971/3971 ━━━━━━━━━━━━━━━━━━━━ 141s 30ms/step - accuracy: 0.8486 - loss: 0.3454 - val_accuracy: 0.8657 - val_los
s: 0.3072
  Fold 3/3 completed. Validation Accuracy: 0.8657
Parameter Set 4 completed. Average CV Accuracy: 0.8618

Best Params: {'lstm_units': 128, 'dropout_rate': 0.3}, Best CV Accuracy: 0.8652
```

In [36]:

```python
# Train final model with real-time progress
final_model = create_model(**best_params)
print("\nTraining final model with best parameters...")
history = final_model.fit(X_train, y_train, epochs=3, batch_size=64, verbose=1,  # Match batch size, keep verbo
                          validation_data=(X_val, y_val),
                          callbacks=[EarlyStopping(monitor='val_loss', patience=3, restore_best_weights=True)])
print("\nFinal Model Summary:")
final_model.summary()
```

```
Training final model with best parameters...
Epoch 1/3
5956/5956 ━━━━━━━━━━━━━━━━━━━━ 171s 28ms/step - accuracy: 0.7349 - loss: 0.5115 - val_accuracy: 0.8472 - val_los
s: 0.3378
Epoch 2/3
5956/5956 ━━━━━━━━━━━━━━━━━━━━ 168s 28ms/step - accuracy: 0.8542 - loss: 0.3279 - val_accuracy: 0.8660 - val_los
s: 0.3024
Epoch 3/3
5956/5956 ━━━━━━━━━━━━━━━━━━━━ 170s 28ms/step - accuracy: 0.8824 - loss: 0.2733 - val_accuracy: 0.8791 - val_los
s: 0.2751

Final Model Summary:
```

Model: "functional_13"

Layer (type)	Output Shape	Param #
input_layer_13 (InputLayer)	(None, 200)	0
embedding_13 (Embedding)	(None, 200, 100)	2,000,000
bidirectional_13 (Bidirectional)	(None, 200, 256)	234,496
custom_attention_13 (CustomAttention)	(None, 256)	456
dropout_26 (Dropout)	(None, 256)	0
dense_26 (Dense)	(None, 64)	16,448
dropout_27 (Dropout)	(None, 64)	0
dense_27 (Dense)	(None, 1)	65

Total params: 2,754,397 (10.51 MB)

Trainable params: 251,465 (982.29 KB)

Non-trainable params: 2,000,000 (7.63 MB)

Optimizer params: 502,932 (1.92 MB)

In [37]:

```python
# Predict on test set
y_pred_prob = final_model.predict(X_test)
y_pred = (y_pred_prob > 0.5).astype(int)
```

```
3971/3971 ━━━━━━━━━━━━━━━━━━━━ 30s 7ms/step
```

Evaluations

In [38]:

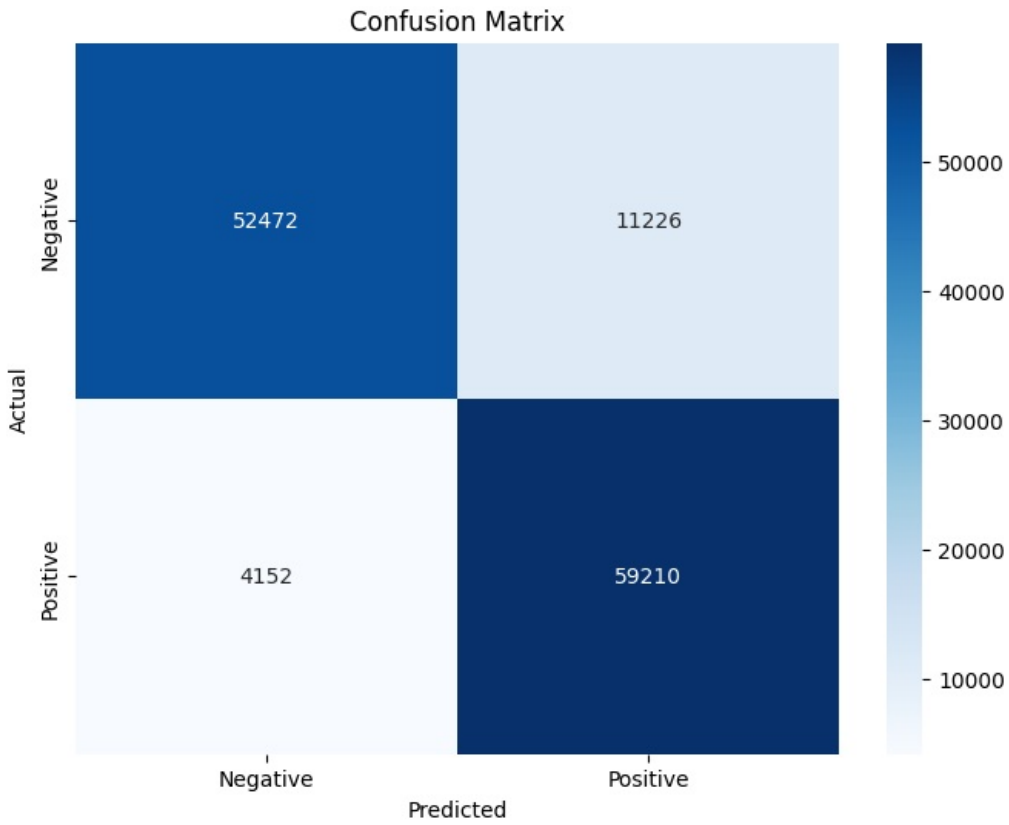
```python
# Metrics
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
print("\nTest Set Evaluation Metrics:")
print(f"Accuracy: {accuracy:.4f}")
print(f"Precision: {precision:.4f}")
print(f"Recall: {recall:.4f}")
```

```
print(f"F1-Score: {f1:.4f}")
```

```
Test Set Evaluation Metrics:
Accuracy: 0.8790
Precision: 0.8406
Recall: 0.9345
F1-Score: 0.8851
```

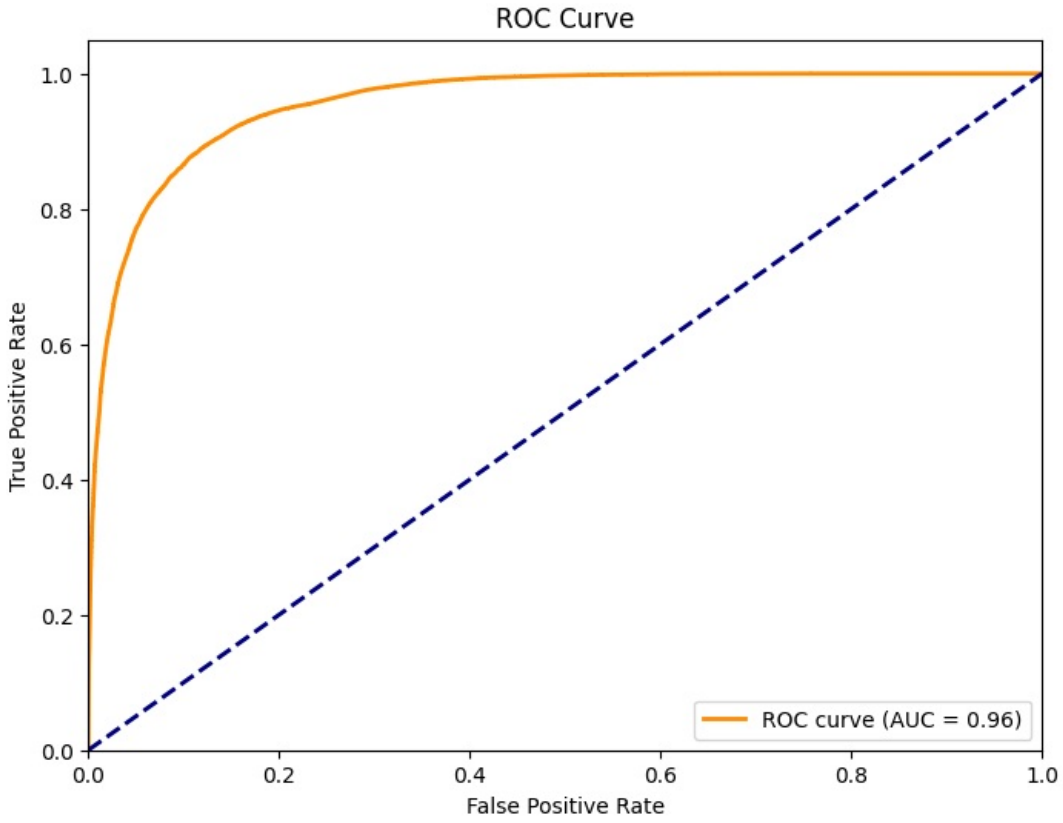
In [39]:

```
# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(8, 6))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=['Negative', 'Positive'], yticklabels=['Negative', 'Positive'])
plt.title('Confusion Matrix')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.show()
```

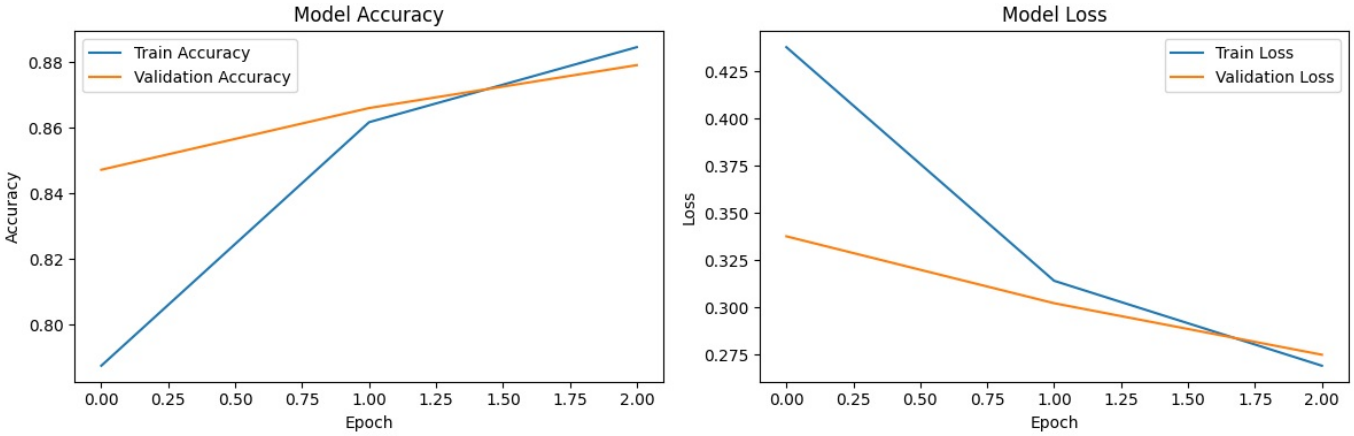


In [40]:

```
# ROC Curve
fpr, tpr, _ = roc_curve(y_test, y_pred_prob)
roc_auc = auc(fpr, tpr)
plt.figure(figsize=(8, 6))
plt.plot(fpr, tpr, color='darkorange', lw=2, label=f'ROC curve (AUC = {roc_auc:.2f})')
plt.plot([0, 1], [0, 1], color='navy', lw=2, linestyle='--')
plt.xlim([0.0, 1.0])
plt.ylim([0.0, 1.05])
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('ROC Curve')
plt.legend(loc="lower right")
plt.show()
```



In [41]:
```
# Learning Curves
plt.figure(figsize=(12, 4))
plt.subplot(1, 2, 1)
plt.plot(history.history['accuracy'], label='Train Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
plt.title('Model Accuracy')
plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.legend()
plt.subplot(1, 2, 2)
plt.plot(history.history['loss'], label='Train Loss')
plt.plot(history.history['val_loss'], label='Validation Loss')
plt.title('Model Loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.legend()
plt.tight_layout()
plt.show()
```



In [2]:
```
# Insights
print("\nInsights:")
print("Strengths:")
print("- Data augmentation and SMOTE ensure robust, balanced training data.")
print("- Bidirectional LSTM with custom Attention captures nuanced sentiment.")
print("- Gradient clipping and k-fold CV enhance training stability and reliability.")
print("Limitations:")
print("- High computational cost due to Bidirectional LSTM and augmentation.")
print("- Synthetic data (SMOTE, augmentation) may introduce subtle biases.")
print("Improvements:")
print("- Use contextual embeddings (e.g., BERT) for even better performance.")
print("- Implement ensemble methods for improved robustness.")
print("- Optimize max_len dynamically based on review length distribution.")
```

```
Insights:
Strengths:
- Data augmentation and SMOTE ensure robust, balanced training data.
- Bidirectional LSTM with custom Attention captures nuanced sentiment.
- Gradient clipping and k-fold CV enhance training stability and reliability.
Limitations:
- High computational cost due to Bidirectional LSTM and augmentation.
- Synthetic data (SMOTE, augmentation) may introduce subtle biases.
Improvements:
- Use contextual embeddings (e.g., BERT) for even better performance.
- Implement ensemble methods for improved robustness.
- Optimize max_len dynamically based on review length distribution.
```

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